ARCH TECHNOLOGIES
Data Science **Internship Report**

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Customer Segmentation Using K-Means Clustering

Table of Contents

1. Introduction

Customer segmentation is an important technique used by businesses to divide customers into groups based on similar characteristics and behavior. It helps companies understand customer purchasing habits and improve marketing strategies.

In this project, K-Means Clustering was used to segment mall customers based on their Annual Income and Spending Score.

2. Project Objective

The objectives of this project are:

- To analyze customer purchasing behavior.
- To perform Exploratory Data Analysis (EDA).
- To identify customer groups using clustering.
- To visualize customer segments.
- To understand cluster characteristics for business insights.

3. Tools & Technologies Used

Tool / Library	Purpose
Python	Programming Language
Pandas	Data Handling
NumPy	Numerical Operations
Matplotlib	Data Visualization
Seaborn	Statistical Visualization
Scikit-learn	Machine Learning
Jupyter Notebook	Development Environment

4. Dataset Information

Dataset Name

Mall Customers Dataset

Dataset Source

Kaggle

Dataset Overview

The dataset contains customer information collected from a mall. It is commonly used for customer segmentation and clustering projects.

5. Exploratory Data Analysis (EDA)

EDA was performed to understand the dataset structure and customer behavior.

EDA Steps Performed

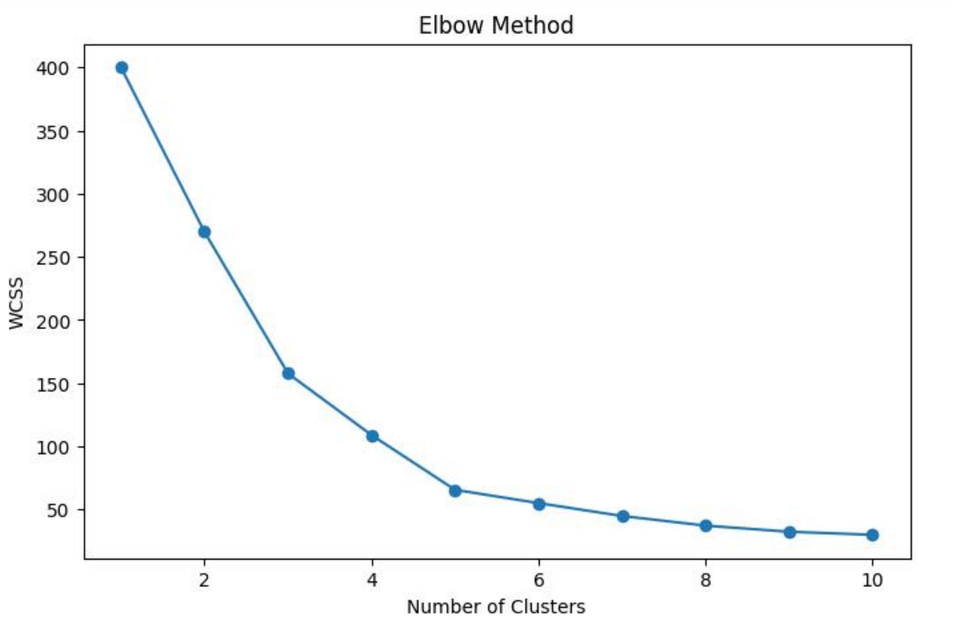
- Displayed first 5 rows.
- Checked dataset information.
- Checked missing values.
- Selected important features.
- Applied feature scaling.
- Used Elbow Method for optimal clusters.

6. Data Visualization

Several visualizations were created to better understand customer patterns and clusters.

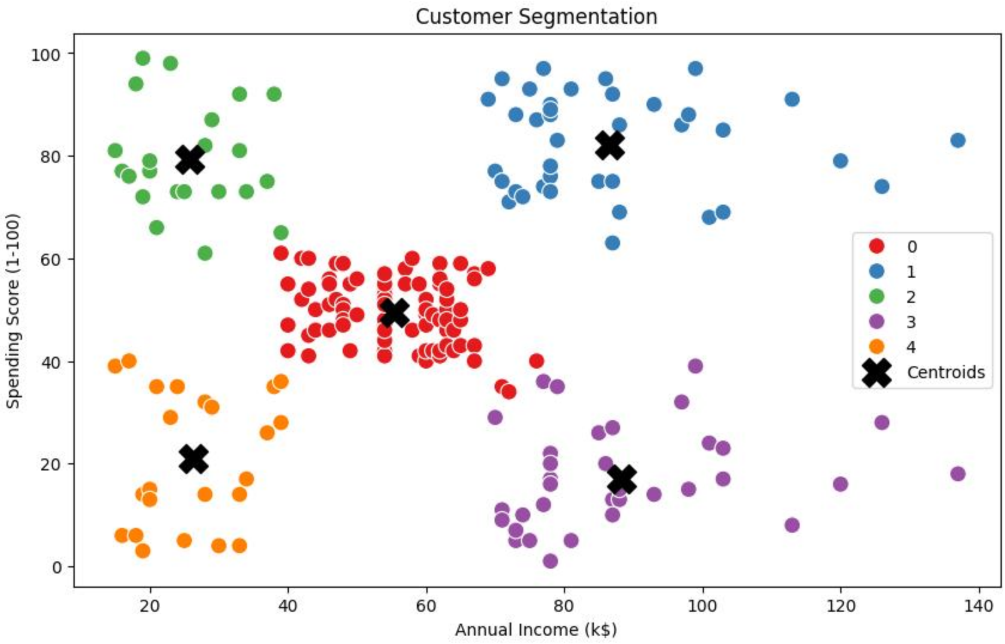
Visualization 1: Elbow Method Graph

The Elbow Method was used to determine the optimal number of clusters.



Visualization 2: Customer Segmentation Scatter Plot

This graph shows customer groups based on income and spending score.



7. Machine Learning Model

Model Used

K-Means Clustering

Why K-Means?

K-Means is an unsupervised Machine Learning algorithm used to group similar data points into clusters. It is simple, efficient, and widely used for customer segmentation tasks.

8. Cluster Analysis

After applying K-Means clustering, customers were divided into 5 groups.

Cluster Insights

- Some customers had high income and high spending behavior.
- Some customers had high income but low spending behavior.
- Some customers had low income and low spending scores.
- Clustering helps businesses target customers effectively.

Code Implementation

Import Libraries

```
# Step 1: Import Libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.cluster import KMeans
from sklearn.preprocessing import StandardScaler
```

Load Dataset

```
# Step 2: Load Dataset
df = pd.read_csv("Mall_Customers.csv")
```

Select Features

```
# Step 4: Select Features
X = df[['Annual Income (k$)', 'Spending Score (1-100)']]
```

Elbow Method

```
# Step 6: Elbow Method
wcss = []

for i in range(1, 11):

    kmeans = KMeans(
        n_clusters=i,
        init='k-means++',
        random_state=42,
        n_init=10
    )

    kmeans.fit(X_scaled)

    wcss.append(kmeans.inertia_)
```

Apply K-Means Clustering

```
# Step 8: Apply K-Means
kmeans = KMeans(
    n_clusters=5,
    init='k-means++',
    random_state=42,
    n_init=10
)

df['Cluster'] = kmeans.fit_predict(X_scaled)
```

Visualization

```
# Step 10: Visualization
plt.figure(figsize=(10,6))

sns.scatterplot(
    x=df['Annual Income (k$)'],
    y=df['Spending Score (1-100)'],
    hue=df['Cluster'],
    palette='Set1',
    s=100
)
```

10. Results & Findings

Findings

- Customers were successfully grouped into clusters.
- Spending behavior varies significantly among customer groups.
- High-income customers do not always spend more.
- K-Means clustering effectively identified hidden customer patterns.

11. Conclusion

This project successfully implemented Customer Segmentation using K-Means Clustering. Different customer groups were identified based on annual income and spending behavior.

The project improved practical understanding of:

- Data preprocessing
- Feature scaling
- Clustering algorithms
- Data visualization
- Business analytics

Movie Rating Prediction

Table of Contents

1. Introduction

Movie recommendation systems and rating prediction models are widely used in modern entertainment platforms such as Netflix, Amazon Prime Video, and YouTube. These systems help users discover movies based on their interests and previous ratings.

This project focuses on building a simple Movie Rating Prediction System using Machine Learning. The model predicts movie ratings based on user IDs and movie IDs using Linear Regression.

2. Project Objective

The main objectives of this project are:

- To understand recommendation system basics.
- To perform Exploratory Data Analysis (EDA).
- To visualize movie rating patterns.
- To train a Machine Learning model for rating prediction.
- To evaluate the performance of the model.
- To save prediction results for future analysis.

3. Tools & Technologies Used

Tool / Library	Purpose
Python	Programming Language
Pandas	Data Handling
Matplotlib	Data Visualization
Seaborn	Statistical Visualization
Scikit-learn	Machine Learning
Jupyter Notebook / VS Code	Development Environment

4. Dataset Information

Dataset Name

MovieLens Latest Small Dataset

Dataset Source

Kaggle

Dataset Overview

The dataset contains movie ratings provided by multiple users for different movies. It is commonly used for recommendation system projects.

5. Exploratory Data Analysis (EDA)

EDA was performed to understand the structure and behavior of the dataset.

EDA Steps Performed

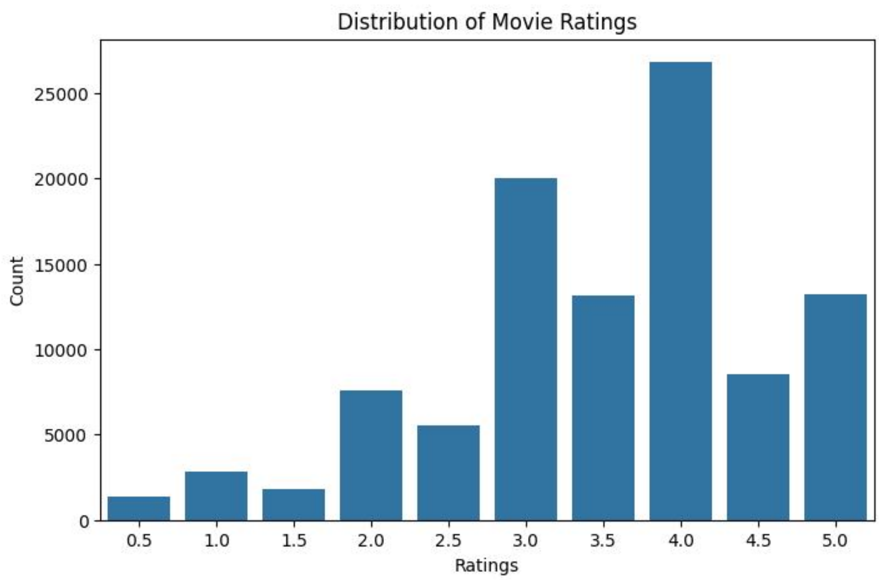
- Displayed first 5 rows of dataset.
- Checked dataset shape.
- Analyzed missing values.
- Viewed statistical summary.
- Observed rating distributions.
- Identified most active users.
- Identified most rated movies.
- Generated correlation heatmap.

6. Data Visualization

Several visualizations were created for better understanding of the dataset.

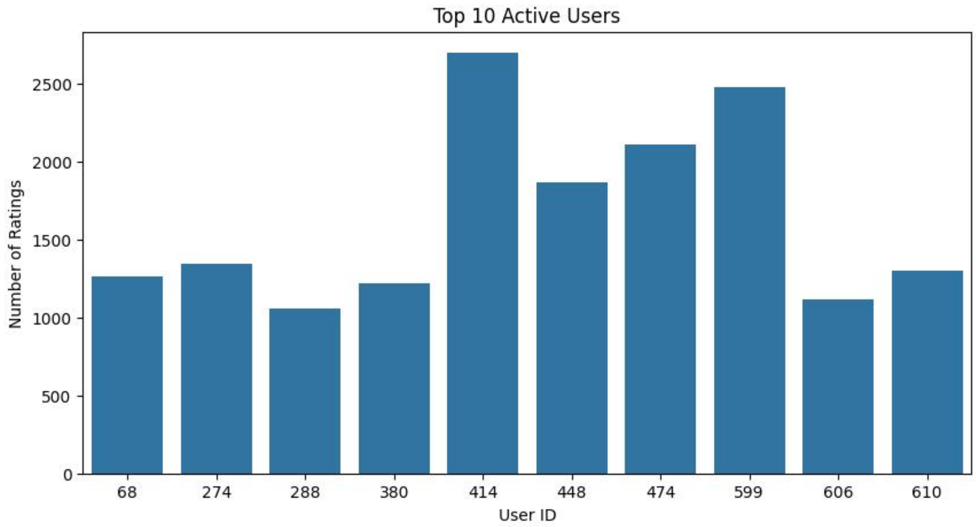
Visualization 1: Distribution of Movie Ratings

This graph shows how frequently each rating appears in the dataset.



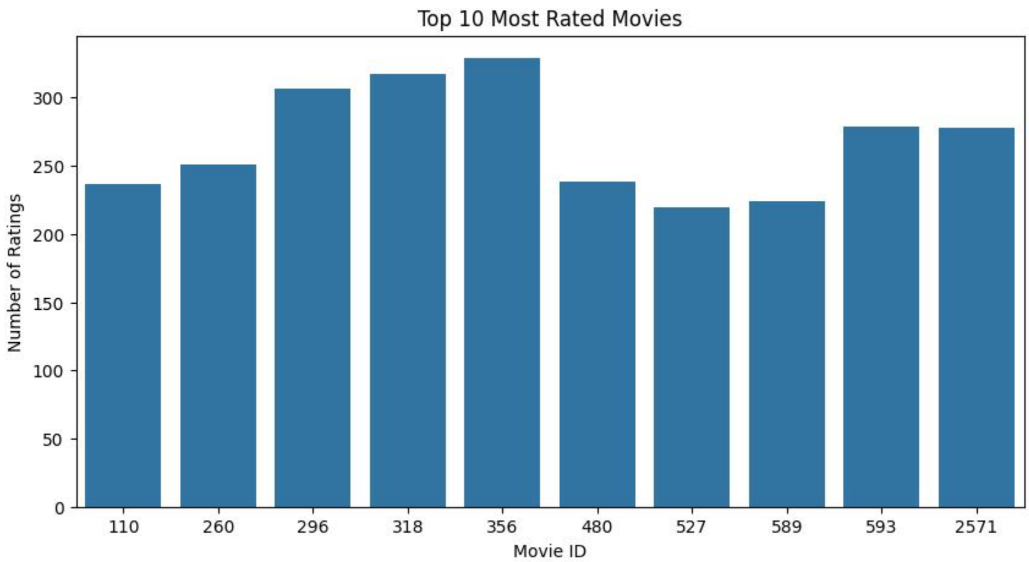
Visualization 2: Top 10 Active Users

This graph represents users who provided the highest number of ratings.



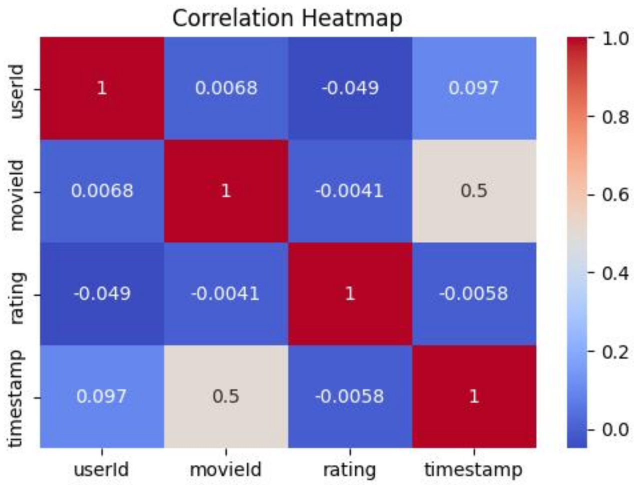
Visualization 3: Top 10 Most Rated Movies

This graph displays movies receiving the highest number of ratings.



Visualization 4: Correlation Heatmap

This heatmap shows relationships between numerical columns.



7. Machine Learning Model

Model Used

Linear Regression

Why Linear Regression?

Linear Regression is a simple and efficient Machine Learning algorithm used for predicting continuous numerical values. In this project, it predicts movie ratings based on user and movie information.

8. Model Evaluation

The model was evaluated using the following metrics:

```
===== Model Evaluation =====
MAE : 0.835
MSE : 1.098
R2 Score : 0.002
```

Code Implementation

Import Libraries

```
# Step 1: Import Libraries
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression

from sklearn.metrics import (
    mean_absolute_error,
    mean_squared_error,
    r2_score
)
```

Load Dataset

```
# ==================================
# Step 2: Load Dataset
# ==================================

df = pd.read_csv("ratings.csv")
```

Feature Selection

```
# ==================================
# Step 5: Feature Selection
# ==================================

X = df[['userId', 'movieId']]
y = df['rating']
```

Train Test Split

```
# ==================================
# Step 6: Train Test Split
# ==================================

X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.2,
    random_state=42
)
```

Train Linear Regression Model

```
# ==================================
# Step 7: Train Model
# ==================================

model = LinearRegression()

model.fit(X_train, y_train)
```

Make Predictions

```
# ==================================
# Step 8: Predictions
# ==================================

predictions = model.predict(X_test)
```

Evaluate Model

```
# ==================================
# Step 9: Evaluation
# ==================================

mae = mean_absolute_error(y_test, predictions)
mse = mean_squared_error(y_test, predictions)
r2 = r2_score(y_test, predictions)
```

10. Results & Findings

Findings

- Most users gave ratings between 3 and 5.
- Some users were highly active in rating movies.
- Certain movies received significantly more ratings.
- Linear Regression successfully predicted approximate ratings.
- The project demonstrates the basic workflow of recommendation systems.

11. Conclusion

This project successfully implemented a Movie Rating Prediction System using Machine Learning techniques. Exploratory Data Analysis and visualizations helped understand user behavior and movie rating patterns. A Linear Regression model was trained and evaluated for predicting ratings.

The project improved practical understanding of:

- Data preprocessing
- EDA
- Data visualization
- Machine Learning workflow
- Model evaluation